

Abstract:

Through the use of electrode sensors placed on various parts of the scalp, EEG scans measure the electrical output produced by the neural activity of different regions of the brain. EEG data is notoriously very noisy, as it includes several unrelated and often subconscious functions of the brain, and electrical output as an interface for interpretation is inherently chaotic, often including environmental noise. To interpret these results, a series of 3-class classification models were developed using incrementally sophisticated techniques with the goal of correctly associating regions of interest with associated event labels trained on a PhysioNet dataset with 109 participants. A basic machine learning pipeline, which compressed the data into a one-dimensional series of values, yielded a peak ~45% test accuracy, a single layer and subsequent double layer CNN (convolutional neural network) yielded ~61% and ~64.5-71% (depending on task), while an EEGNet architecture achieved ~64-70% though with a consistent pattern of superior generalization. An ensembled prediction utilizing one instance of either model yielded an accuracy of **74.92%**. Results showed that while the model could successfully differentiate between rest and movement, learning the difference between motor tasks was much more difficult. Though this signal proved to be significantly easier to interpret in instances of executed movement, and surprisingly, EEGNet's parameter-efficient architecture demonstrated superior performance over the CNN when examining Fists/Feet tasks, suggesting it may more effectively generalize the weaker, spatially diffuse signals associated with the deeply imbedded foot motor cortex activity. Though combining predictions of different representations produces the most diverse understanding of the information, yielding the highest percentage of accuracy. .

Introduction:

EEG scans are remarkable as they provide the means to cheaply and holistically examine the brain's neural activity at a millisecond temporal resolution in a non-invasive manner, aspects that make it invaluable for real-time scientific and medical applications as well as research. Though EEG spatial fidelity leaves much to be desired compared to other scanning methods such as fMRI, which measures blood flow within the brain at a very precise resolution despite taking substantially more time to complete. One of the most widely studied EEG applications is motor imagery classification, in which an individual imagines or executes a movement while remaining physically stationary. A successful implementation of this classification aids many fields from assistive technologies for patients with paralysis to neurorehabilitation and human-computer interaction. The goal was to assess to what extent this EEG data could be interpreted, with the hope that techniques such as ensembling, aggregating results from several competent models, and temporal majority voting, wherein predictions across consecutive epochs belonging to the same trial are aggregated to produce a more reliable class estimate, could approach a research-grade accuracy (>80%). This task is accomplished by implementing neural networks classifiers that extract the data, process and optimize its representation, extract relevant features, and use the final abstract representation to train a model which eventually lands on a class identification.

Though this process is easier imagined in theory than executed as there exist many challenges researchers have struggled to solve throughout the years. The most immediate barrier to entry is feature extraction. Neural signals recorded at the scalp exhibit a low signal-to-noise ratio due to physiological artifacts, environmental interference, and variability between individuals. Additionally, signals produced from individual regions often bleed into other nearby channels, further obscuring signal prominence.

Methods:

Gerwin Schalk and his team of researchers conducted an experiment on 109 participants, asking them to perform various motor tasks/imagery while under an EEG setup. This task was performed under three repetitions of four two-minute task conditions:

Run	Task
R01	Baseline, eyes open
R02	Baseline, eyes closed
R03, R07, R11	Execute left vs right fist
R04, R08, R12	Imagine left vs right fist
R05, R09, R13	Execute both fists vs both feet
R06, R10, R14	Imagine both fists vs both feet

Several methods were used to extract clarity from the raw EEG data and models were implemented using various architectures to discover a superior method for interpreting these signals:

Basic Preprocessing:

Initial preprocessing of the EEG data involves band-pass filtering and separating data into segmented samples to be loaded into the model, a process known as epoching. Motor movement and motor imagery are primarily associated with activity in the μ (8–13 Hz) and β (13–30 Hz) frequency bands. Consequently, a band-pass filter of 8–30 Hz was applied to suppress unrelated neural activity and environmental noise while preserving the frequencies most relevant to motor tasks.

After filtering, the data is epoched and collected. In this dataset, EEG temporal resolution represents neural activity at a rate of 160Hz per second. Continuous EEG recordings were

segmented into epochs extending from 0.5 seconds before each event onset to 2.0 seconds after the event. The 0.5 second pre-stimulus interval was used for baseline correction, removing slow voltage drift and inter-trial variability by subtracting the mean signal immediately preceding each task. This yielded ~320 temporal samples for every epoch.

Multilayer Perceptron (MLP) Baseline:

An MLP was the initial model implemented to interpret the EEG data, a method that distinctly lacks any feature extraction. This architecture takes only the filtered raw data directly from the initial preprocessing step and feeds it into a neural network by flattening all the spatial-temporal data into a one-dimensional vector of values. This was done in hope that there existed some learnable signal even within a flattening of the spatial and temporal structure into a single one-dimensional feature vector, thereby demonstrating a need to improve the architecture, as this baseline was to be integrated into every subsequent implementation with emphasis on additional intermediate feature extraction steps. The input layer consisted of approximately 64×321 features, corresponding to 64 EEG channels sampled over the duration of each epoch, and an output size of 3, representing the number of possible event labels. A hidden size of 16 was selected to minimize over-fitting, and epochs were loaded into batches of 35.

Convolutional Neural Networks:

To increase performance, a convolutional stage was implemented between preprocessing and input into the final fully connected network. A convolutional layer takes the initial data and learns several localized spatiotemporal features by sliding multiple filters across the temporal dimension while simultaneously integrating information from all EEG channels. During training, these filters automatically adapt to detect patterns that best distinguish the motor tasks, aggregating values into fewer channels of cleaner information. The network progressively extracts increasingly informative representations before pooling and flattening into an appropriate format for classification by the final fully connected layer.

This process may be repeated multiple times until an optimal test-accuracy is achieved, as there exists a point where additional abstraction becomes redundant or worsens results. In its initial layer, the CNN implemented in this model learns 32 spatial patterns spanning all 64 channels. Batch normalization was applied to stabilize training, followed by max pooling to reduce dimensionality while preserving relevant features. Subsequent layers perform essentially the same operation on higher level abstracted representations, with the only difference being in the dimensionality, differing as 64 channels are compressed into increasingly fewer channels. This process allows the model to learn progressively more complicated patterns and consequently provide better classification.

EEGNet Architecture:

EEGNet is a compact convolutional neural network specifically designed for EEG classification. The architecture differs from traditional CNN implementations by prioritizing efficient feature

representation while utilizing significantly fewer trainable parameters. This is achieved by separating temporal and spatial filtering into separate stages of abstraction, rather than learning them simultaneously. Consequently, EEGNet has demonstrated strong performance across a variety of brain-computer interface applications, particularly when limited training data are available, where smaller architectures often generalize more effectively than conventional CNNs.

EEGNet is implemented through two primary stages: a depthwise stage and a pointwise stage. Following an initial temporal convolution that learns frequency-dependent activity independently for each EEG channel, the depthwise convolution learns spatial filters across all 64 electrodes for each temporal feature independently. The outputs of the temporal convolution are referred to as feature maps, and for every map, two spatial filters of information are extracted, producing compact spatiotemporal representations of the EEG signal. So, spatial filters compress every channel into just one unique feature. Each resulting feature therefore captures both what temporal pattern was detected and where it occurred across the scalp.

The pointwise step refines this feature in every map, compressing values across temporal windows, before combining all the data into one robust representation, producing a compact yet expressive representation that is passed to the fully connected classifier.

Common Spatial Patterns:

CSP is a feature processing method that attempts to make differentiating signals within data more pronounced. It does this by identifying spatial patterns in the EEG channels whose activity differs most strongly between two classes, then projecting the original signals through these learned spatial filters to emphasize class-specific activity while suppressing patterns shared between classes. It inherently works as a binary classifier, meaning, within the context of a three-class problem, its use must be strategically implemented. Our use case was to use CSP in hopes of better discrimination between motor movement/imagery, ignoring the rest event, as that's where model performance would prove to struggle.

Because traditional CSP summarizes each spatial component by its temporal variance, it removes the temporal structure required by convolutional neural networks. To preserve temporal information, an alternative implementation was investigated in which the learned CSP spatial filters were applied directly to the EEG recordings while retaining the time dimension. This allowed the CNN to operate on spatially enhanced signals rather than variance-based features.

Results:

Model performance:

Trial:	CNN	EEGNet
Imagined L/R:R04, R08, R12	64.93%	62.36%

Executed L/R:R03, R07, R11	71.14%	68.88%
Exe Fists/Feet: R05, R09, R13	67.67%	69.98%
Imagined Fists/Feet: R06, R10, R14	57.75%	58.70%

Multilayer Perceptron (MLP) Baseline:

Implementing a basic multilayer perceptron without any sort of substantial feature processing plainly illustrates the limits of working with raw EEG Data. A model trained without a feature extraction pipeline yields ~**45%** accuracy, using task condition R04, R08, R12, within just three to four epochs before gradually declining to ~**40%** as training accuracy quickly climbs to nearly **100%**. This result is due to dramatic overfitting, effectively a difference between memorization and understanding, which while demonstrating some learned accuracy, fails to provide reliable classification, practical application, or potential research outcome. This however gives cause for a more developed feature extraction/representation.

Convolutional Neural Networks:

Introducing convolutional feature extraction significantly improved classification performance over the MLP baseline. Through the preservation of the spatial and temporal structure of the EEG recordings, the CNN learned spatiotemporal features that generalized more effectively to new, unseen data. A single convolutional layer improved test accuracy to approximately **61%**, while the addition of a second convolutional layer increased performance to roughly **64–71%**, depending on the motor task. Though a third convolutional layer failed to provide further improvement in accuracy, suggesting that the information available had been thoroughly extracted and that performance is limited by the quality of the raw EEG signal rather than model complexity.

EEGNet Architecture:

EEGNet achieved performance comparable to the best-performing CNN while utilizing substantially fewer trainable parameters. Although overall classification accuracy was similar, EEGNet consistently demonstrated improved generalization, particularly on the executed **fists versus feet** task. This suggests that separating temporal filtering, spatial filtering, and feature integration through depthwise and pointwise convolutions provides a more parameter-efficient representation of EEG activity without sacrificing predictive performance. These findings support EEGNet's suitability for EEG classification tasks where training data is relatively limited.

Common Spatial Patterns:

A novel implementation of CSP did demonstrate a significant improvement in accuracy over the base MLP in distinguishing between motor signals, though it failed to surpass the accuracy already yielded by CNN and EEGNet architectures. On the two-class problem, CSP achieves

roughly 67-70% accuracy, though this figure is not directly comparable to the figures yielded by the three-class convolutional architectures. The relevant discussion asks, in the instance motor imagery is detected, how does CSP compare to other architectures in its predictions. Analysis of the confusion matrices finds that raw CNN accuracy yields roughly 77%, on the condition motor signal is already detected. This suggests that the spatial feature processing introduced by CSP did not provide a clear advantage over the spatial-temporal representations already learned by the convolutional architectures.

Architecture changes:

Changes Tested	Result	Interpretation
Kernel size (1,10) → (64,10)	Initially observed improvement from 52.59% accuracy to 60.42%	This change offered the largest single improvement as shifting the kernel size expanded the model's view from 1 channel to all 64 simultaneously per time window
Dropout (p=0.5 → 0.25 → 0.1)	Stabilized noisy test accuracy, peak ~61%	Reduced the variance in test accuracy results, despite maintaining the same peak. Also regulated the divergence in training accuracy and testing accuracy as the model learned
Doubling filter count (32→64, 64→128)	~64.58% (no improvement)	Ruled out width as the bottleneck. Wider network reached similar accuracy but required more passes and had a worse train/test gap (94% train vs 65% test), indicating pure overfitting with no real gain.
Weight decay (L2, 1e-4)	~63.82% (no improvement)	Ruled out additional regularization as the crux
3rd conv layer	~57.6% (worse)	Ruled out depth (at this scale). Results indicated increased capacity to memorize (94.89% train) without improving generalization (57.62% test). With ~9,000 total epochs, a

		3-layer CNN likely needs more data to justify the added complexity.
Removing 8-30 Hz bandpass filter	64.33% (no meaningful change)	The model learns despite a filtered range, indicating that filtering neither suppresses nor highlights any useful information for the model.
Class-weighted loss (for imbalance)	61.89% (worse)	Class imbalance (45/23/22 split in early single-subject testing) wasn't severe enough to justify the precision/recall trade-off.

Discussion:

Diagnostics:

Best Test Accuracy: 71.14% - (R03, R07, R11)

Table 1. Classification Performance:

Class	Precision	Recall	F1-Score	Support
Rest	0.80	0.86	0.83	920
Left Fist	0.65	0.66	0.65	495
Right Fist	0.67	0.58	0.62	491
Overall Accuracy			0.73	1906
Macro Average	0.71	0.70	0.70	1906
Weighted Average	0.73	0.73	0.73	1906

Table 2. Confusion Matrix:

True Class	Predicted Rest	Predicted Left	Predicted Right
Rest	787	81	52
Left Fist	83	325	87
Right Fist	113	94	284

The model most reliably identified the resting state, achieving an F1-score of 0.83 and correctly classifying 86% of resting epochs. Classification of left and right motor imagery proved substantially more difficult, with the largest source of error arising from confusion between the two movement classes rather than between movement and rest.

Findings:

A basic MLP rapidly overfit the EEG data, demonstrating that a raw flattened EEG representation does not offer sufficient feature representation for any reliable classification, which gives cause to implement feature extraction. Introducing convolutional layers substantially improved classification by preserving spatial and temporal structure, though increasing CNN depth beyond two layers worsened accuracy, suggesting additional architecture complexity would only result in diminishing returns. Subsequent variations in architecture include an implementation of EEGNet, which performed comparably to conventional CNN's while using significantly fewer trainable parameters, displaying superior parameter efficiency. This fundamental change in architecture also yields a more generalized model of understanding, as less parameters inherently discourage learned patterns of overfitting. After evaluating multiple architectures, preprocessing and EEG signal quality appeared to be the primary factors limiting classification performance rather than network complexity.

Through analysis of the diagnostic information, the primary source of classification error was found to be distinguishing between motor tasks rather than separating movement from rest. Executed motor tasks consistently achieved higher classification accuracy than imagined motor tasks, reflecting a stronger, more pronounced neural signal. This result is expected, as motor movement is known to produce greater neural activity than motor imagery. Models consistently proved themselves to be more capable of distinguishing left-hand versus right-hand trials than fists versus feet trials. This may be a consequence of neural activity associated with foot motor movement being more deeply imbedded within the brain, making it more difficult for scalp EEG sensors to detect. Left-hand motor activity was also consistently easier to classify than right-hand motor activity, across multiple model architectures, suggesting some underlying asymmetry in the dataset or neural signals. A novel implementation of CSP-based preprocessing did not outperform learned CNN feature extraction, indicating that fixed spatial transformations reveal limited additional information compared to CNN-based learned spatial representations.

Ensembling :

Task: R03, R07, R11

Best Ensembled Test Accuracy: 74.92%

Ensemble Composition	Test Accuracy
3 CNN + 2 EEGNet	73.50%
2 CNN + 2 EEGNet	74.82%
2 CNN + 1 EEGNet	74.50%
1 CNN + 2 EEGNet	74.71%

1 CNN + 1 EEGNet	74.92%
3CNN	72.93%
2CNN	74.13%
2EEG	69.78%

Combining CNN and EEGNet architectures consistently improved performance compared to ensembles consisting of only CNN models or EEGNet models, demonstrating that architectural diversity provided more useful predictive variation than simply increasing the number of models. This result is to be expected as different architectures learn significantly different models of the information, resulting in a more diverse understanding of the problem, whereas the same architecture is more likely to make predictions while making similar errors. The strongest ensemble consisted of only one CNN and one EEGNet model, achieving **74.92% accuracy**, a ~3.8 percentage point improvement over the best individual model, suggesting that these architectures learned complementary representations of the EEG signal. Increasing ensemble size beyond two models demonstrated diminishing returns, with every larger ensemble failing to surpass smaller configurations. The strictly double-CNN ensemble performed competitively with combined ensembles, while EEGNet-only ensembles showed weaker performance, suggesting that this specific EEGNet implementation may have captured less discriminative information. Negative returns from additional models suggest that ensemble performance was primarily constrained by shared difficulties in classification, rather than model variance. These results indicated that combining distinct feature extraction strategies is likely more valuable than either aggregating many models with similar learned representations or increases in depth within individual architectures.

Conclusion:

This study examined the extent to which motor-imagery could be predicted from EEG scans within this specific data set, with the implementation of progressively more sophisticated architectures and a variety of techniques. The results yielded by the optimal strategy indicate that meaningful classification is possible, and it is reasonable to assume this is the case beyond just this data set. A basic multilayer perceptron trained directly on flattened EEG data exhibited severe overfitting and limited generalization, while convolutional architectures substantially improved performance by preserving and learning from the spatial and temporal structure of the signal. The strongest individual models achieved approximately 70–71% test accuracy, while combining predictions from a CNN and EEGNet increased accuracy to 74.92%.

These results demonstrate that EEG scans, while noisy and difficult to design feature extraction around, contain sufficient information from which to produce models and make meaningfully

accurate predictions. However, there is a clear ceiling as to what extent this is possible, visible through examination of the model's difficulty in differentiating between motor tasks. Architecture diversity proved to be a significant avenue through which greater accuracy may be achieved, as ensembling combines conclusions of models with differing strengths and weaknesses. Attempts were made to aid the difficulty in differentiation between motor tasks, such as the implementation of a binary CSP pipeline, though this strategy yielded weaker feature extraction than what convolutional networks already achieved. Despite these difficulties, a significant degree of success was found concerning the initial question of to what extent motor imagery could be predicted. Many fundamental aspects relating to the data and the obtainability of neural activity were uncovered, while also highlighting the inherent variability and information limitations that constrain classification accuracy.

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